Mahfood Balfaqeeh

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Education

York University — Toronto, ON
Honours Bachelor of Science in Computer Science, Dean's Honour Roll — *Sep. 2022 – Dec. 2027 (Expected)*

Experience

Software Developer Intern — May 2026 – Present
IBM Skills Network — *Markham, ON*

- Re-architected AI Classroom chat by caching session context in **Redis** and removing **LangChain** as an abstraction layer, cutting response times from **20–25s to 2–3s** (**10x faster**)
- Added prompt caching for system prompts, chat history, and tool calls, and restructured context retrieval to skip unneeded model calls, speeding up responses by **55%+**
- Instrumented the **LangGraph** AI assistant with **OpenTelemetry**, tracing every message and chat with tokens used, tokens cached, and time per chain, which exposed the bottlenecks and inefficiencies that drove the **55%+** speedup
- Built a **NestJS** routing layer unifying **40+ LLMs** from **OpenAI, Anthropic, watsonx.ai, Google, and Together AI** with automatic failover and outage monitoring, powering model experimentation in AI Classroom
- Built a service that feeds live browser state into the AI assistant's context, enabling answers grounded in what users are currently doing

Software Developer Intern — Sep. 2025 – Apr. 2026
IBM Skills Network — *Markham, ON*

- Refactored **React/TypeScript** player state, resolution encoding, and video sourcing, cutting streaming costs by **∼$20K/month**
- Migrated the internal support platform from **Next.js** to **Vite** with statically served pages instead of SSR, first locking in existing behavior with an **end-to-end Playwright** suite so the rewrite shipped without regressions
- Removed the need to run two servers in CI by serving static pages, avoiding extra runner capacity and shrinking the **Kubernetes** pods by half, while cutting test runs by **15 min** and builds by **3 min**
- Extended the **CI/CD** release pipeline to run E2E tests against **Docker** images before release, adding a pluggable hook for inserting custom test actions mid-workflow without modifying the shared central workflow, now adopted by **3 teams**

AI Intern — May 2025 – Aug. 2025
Growth-Hacker — *Montreal, QC*

- Automated keyword-ranking tracking by building a **Python** scraping pipeline integrated with the **SEMrush** and **Ahrefs** APIs
- Replaced manual SEO page updates with a web app that connects live analytics data to automated content generation

Projects

NES Emulator | *C, Raylib* — GitHub

- Wrote an NES emulator in **C**, implementing all **151** official **6502** opcodes (**56** instructions, **12** addressing modes) with cycle-counted timing
- Implemented **2C02 PPU** background/sprite rendering, iNES cartridge loading, and **Mapper 000** mapping
- Built a **60 FPS Raylib** frontend with a debugger for CPU registers, disassembly, and instruction/frame stepping

LokaGPT – Character-Level Transformer | *Python, PyTorch* — GitHub

- Trained a **21M-parameter** GPT from scratch in **PyTorch**, adapting nanoGPT's attention, MLP, and LayerNorm blocks
- Built a char-level tokenizer and **AdamW** training loop with LR warmup/decay over a **98K-character** Q&A corpus

Technical Skills

Languages: TypeScript, JavaScript, Python, C/C++, C#, Java, SQL, Bash, Ruby, Lua
Frameworks: React, NestJS, Node.js, Flask, Tailwind CSS, Ruby on Rails, LangGraph
Developer Tools: Git, Docker, Kubernetes, AWS, Redis, Linux, CI/CD, Playwright, OpenTelemetry, Vite
Libraries: PyTorch, OpenCV, Ultralytics YOLO, Raylib